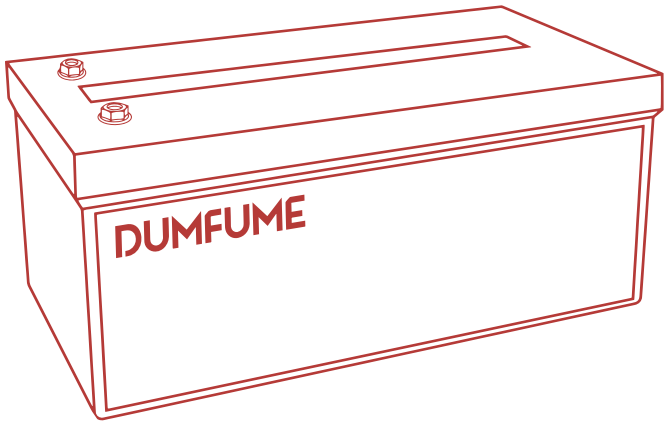
DUMFUME

LiFePO4

LITHUM IRON PHOSPHATE

User Manual



38.4V 100Ah
51.2V 100Ah

Website:www.dumfume.cn
Provider:Shenzhen Dumfume Commerce Co.,Ltd

38.4V 100Ah
51.2V 100Ah

PRODUCT MANUAL

Lithium Iron phosphate
(LiFePO4)Battery

CAUTION

- Flammable/explosive hazard.
- Avoid mechanical impact.
- Do not short circuit.
- Do not crush.
- Do not disassemble.
- Do not incinerate.
- Do not heat above 149°F/65°C.
- Do not immerse the battery in water.
- Do not reverse the polarity of the battery and charger.
- Use the appropriate tools when handling the battery.
- When connecting batteries in parallel, refrain from using batteries of other brands or types, as the Battery Management System (BMS) may not be compatible.
- Keep the battery away from fire, hazardous materials and other dangerous substances.

⚠ WARNING

BATTERY DISPOSAL

- The electrodes of the waste battery should be wrapped with insulating paper to prevent fire and explosion.

PROHIBITION OF USING IN BELOW PLACES

- Prohibited use in environments with strong static electricity or magnetic fields. Such exposure may compromise the battery's safety protection circuitry, creating potential safety hazards.

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PRODUCT OVERVIEW

Key Features

1. Lithium Iron Phosphate (LiFePO4) chemistry provides exceptional stability and consistent performance.
2. Advanced Battery Management System (BMS) ensures product safety and extended lifespan.
3. With powerful output, it can be perfectly compatible with the mainstream golf carts currently on the market.
4. With standard industrial dimensions, it can be perfectly integrated into the battery compartments of most vehicle models.
5. It comes with a built-in intelligent coulomb meter, allowing you to know the accurate power data in real time.
6. It has constant voltage and full usable capacity at any state of charge.
7. It supports fast charging and discharging.
8. Grade UL94 V-0 (Fireproof).

NOTICE BEFORE USING

1. With proper care your Dumfume battery will give you years of trouble-free power.
2. Due to shipping-related regulations, this battery carries about 30%~50% capacity when shipped.
3. Avoid direct contact between the terminals of positive and negative wire connected to the battery, which may cause the battery to short circuit and become unusable.
4. We recommend wrapping the wire terminals with insulating tape during the wiring process.
5. Ensure that the battery is properly wired to the load. Do not reverse the positive and negative connections.
6. Before connecting the load, please make sure the power supported by the battery can meet the load's power demand.
7. The battery is recommended to be stored between 10°C to 35°C (50°F to 95°F) at 50% charge level and recharged every three months for a long-term storage.

⚠ Do not disassemble the battery

- Install the battery upright with post bolt facing up, and it could not be mounted upside down. If you need to mount the battery on its side, please contact our Dumfume after-sales team directly through your order.

BATTERY PARAMETERS

Capacity	51.2V 100 AH	38.4V 100AH
Weight	39.2Kg/86.42Ibs	31.2Kg/68.78Ibs
Dimensions (Nylon Handle)	20.87×10.43×8.66in/ 53.0×26.5×22.0cm	20.87×10.43×8.66in/ 53.0×26.5×22.0cm
Terminal Bolt Size	M8	
Nominal Voltage	51.2V	38.4V
Energy	5120Wh	3840Wh
Charge Method	CC/CV (Constant Current/Voltage)	
Recommended Charge Current	20-50A	
Usable Capacity	100Ah	
Standard Charge Voltage	58.4V	43.8V
Shipping Voltage	51.5-52.8V	38.4-39.6V
Shipping SOC	40%	
Cycle Life	⩾4000 cycles (to 80% capacity)	
Self-Discharge Rate	<3% per month	
Series/Parallel	Max 1S4P	
Communication	Not supported	
Case Material	ABS Plastic	
Pack Certifications	CE/UN38.3/Class 9/IEC62133/RoHS	
Cell Certifications	Cell Certifications	
Storage Temperature	-10~45°C (14~113°F)	

Capacity	51.2V 100 AH	38.4V 100AH
Over -charge Protection Voltage	58.4V	43.8V
Over -charge Recovery Voltage	53.6V	40.2V
Continuous Maximum Charge Current	200A	
Recommended Discharge Current	0-150A	
Continuous Maximum Discharge Current	200A	
Maximum Power	10240W	7680W
Over -discharge Protection Voltage	40V	30V
Over -discharge Recovery Voltage	47.2 V	35.4V
Discharge Over-current rotection	300 A (±80 A)	
Short -circuit Protection Current	1500 A (±50 A)	
Recommended charging voltage	58.4V	43.8V
Charge Low-Temperature Protection	23~41°F	
Charge Low-temperature Protection Recovery Value	32~50°F	
Charge High-temperature Protection	140~158°F	
Charge High-temperature Protection Recovery Value	122 ~140°F	

Capacity	51.2V 100 AH	38.4V 100AH
Discharge Low-temperature Protection	-13~50°F	
Discharge Low-temperature Protection Recovery Value	-4 ~14°F	
Discharge High -temperature Protection	158~176°F	
Discharge High-temperature Protection Recovery Value	140~158°F	
Storage Environment(50% Charge Quantity)	Storage period within three months: Temperature-20~45 °C,Humidity 65±20%RH. Storage period exceeding three months: Temperature 25±3°C, Humidity 65±20%RH.	

Battery Management System (BMS) Specifications

38.4V 100Ah：

State of Charge	Voltage	Technical Notes
100%	40.8V	Full charge voltage (BMS cutoff:43.8/58.4V)
80%	40.2V	High SoC operating range
60%	39.6V	Standard working voltage
40%	39.2V	Mid-discharge voltage
20%	37.8V	Low voltage warning zone
10%	36.6V	Deep discharge limit
0%	30V	Absolute minimum (cell damage risk)

51.2V 100Ah：

State of Charge	Voltage	Technical Notes
100%	54V	Full charge voltage (BMS cutoff:43.8/58.4V)
80%	53.2V	High SoC operating range
60%	52.5V	Standard working voltage
40%	52V	Mid-discharge voltage
20%	50.4V	Low voltage warning zone
10%	48.6V	Deep discharge limit
0%	40V	Absolute minimum (cell damage risk)

*Note：The above data is for reference only. The actual voltage depends on the charger used for charging and other machine parameters

CONNECTION STEPS

STEP 01

- Fully charge each battery individually.

STEP 02

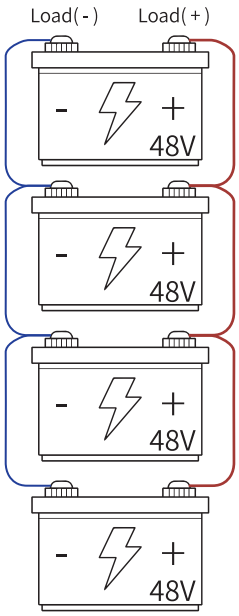
- Wait 15 minutes, then test the voltage. It should be more than 52V.

STEP 03

- Connect the batteries in series or parallel.
  (Taking a 48 V 100 Ah battery as an example).

Parallel Connection

48V 100Ah



- Taking a 48V 100Ah battery as an example, first connect four 48V 100Ah batteries in parallel, and then connect one 48V 100Ah battery in series to form a battery pack with a total voltage of 48 V and a capacity of 400Ah.

BMS FUNCTION DESCRIPTION

STEP 01

Overcharge Protection (>58.4V)

- When any single cell voltage exceeds the preset threshold during charging, the BMS will immediately terminate the charging process.

STEP 02

Over-Discharge Protection (<40V)

- If the battery voltage drops below the threshold during discharge, the BMS will cut off the discharge circuit.
- Auto Recovery: The system will automatically reconnect after 15 seconds when voltage rises above 47.2V.

STEP 03

High-Temperature Protection

- **Charging Protection (65°C/149°F): Charging is disabled when the threshold reached.**
- **Discharging Protection (70°C/158°F): Discharging is disabled when the threshold reached.**

STEP 04

Low-Temperature Protection

- **Charging Protection: Charging is prohibited at ⩽0°C (32°F).**
- **Discharging Protection: Supports discharge down to -20°C (-4°F).**

STEP 05

Over current/Short-Circuit Protection

- **When triggered, the BMS will completely disconnect battery output.**
- **Recovery Procedure:**

  Disconnect ALL battery cables.
  Measure voltage with a multimeter:
  If voltage is >47.2: Reconnect to resume operation.
  If voltage is ⩽40V: please contact our Dumfume after-sales team directly through your order.

Cell Balancing Function Description

If the battery has been disconnected by the BMS due to a protection trigger, please follow these reactivation steps:

Trigger Conditions

- Balancing automatically activates when both conditions are met:

  End of charge cycle (near full capacity).

  When the battery is 51.2V 100Ah, the total voltage exceeds 58.4V (when the battery is 38.4V 100Ah, the total voltage exceeds 43.8V).

Working Principle

- Diverts excess energy from higher-voltage cells via intelligent shunt resistors.
- Equalizes State of Charge (SOC) across all cells (±1% variance).

Technical Advantages

- **Enhanced Pack Performance**
  Reduces voltage deviation to <30mV between cells.

- **Extended Cycle Life**
  Prevents accelerated degradation of overcharged cells.

- **Overcharge Safeguard**
  Limits individual cell voltage to ⩽3.65V during balancing.

CHARGING METHODS

You can charge your lithium iron phosphate batteries whenever you want, just like your cellphone. Unlike lead-acid batteries, lithium iron phosphate batteries do not get damaged if they are left in a partial state of charge, so you don't have to stress about a memory effect, so you don't have to drain them completely before charging.

Battery Charger(mains power)

- The most ideal way to charge a LiFePO4 battery is with a lithium iron phosphate battery charger, as it will be programmed with the appropriate voltage limits. Most lead-acid battery chargers will do the job just fine.
- A 48V lithium battery at 20% capacity will hold voltage around 51.2V, while its lead-acid battery will be approximately 40V at the same capacity. So if you use lead-acid charger to charge your lithium battery, it may not be fully charged.

Solar panel (DC power)

- You can also use the solar panel to charge your DUMFUME LiFePO4 battery, but please make sure to choose a proper controller, which should include the LiFePO4 battery mode or Li-ion battery mode. Both PWM controller and MPPT controller are acceptable.

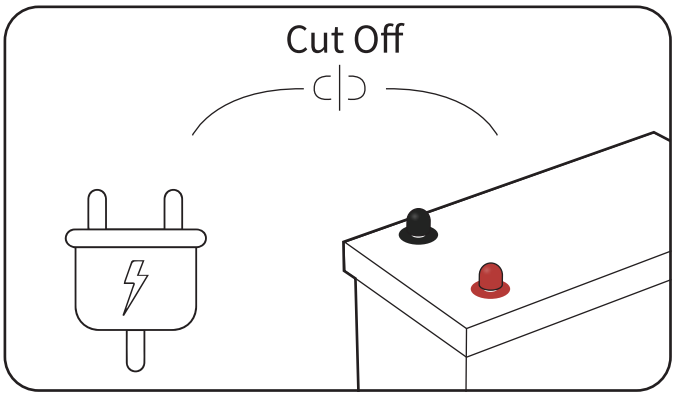
- If you don't have a controller, you can connect the battery directly to the solar panel directly. The BMS inside will protect the battery most of time.But if there is a defect in the battery BMS, the battery will be damaged.
- If you need to test the actual voltage of the battery, please disconnect all the connections to the battery and test its voltage after leaving it aside for over 30 minutes.
- To select a proper charger for your battery, you should choose one that is capable of fully charging the DUMFUME battery. Its rated output/charging voltage should match the LiFePO4 battery's standard charge voltage, which varies depending on the battery types, please refer to the above parameter table for detailed data.

GENERAL STEPS

If the BMS has cut off the battery for protection, follow the below steps to activate it.

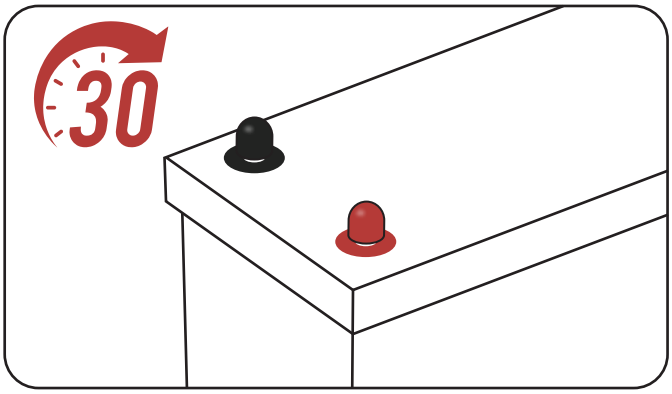
STEP 01

- Cut off all the connections from the battery.



STEP 02

- Leave the battery aside for 30 mins
- Then the battery will automatically recover to normal voltage(>47.2V for 51.2V model / >35.4V for 38.4V model) and can be used after fully charged.



Wide Applications

- RV
- Solar Energy Storage
- Industrial Battery
- Used to Replace 12V Lead-acid Batteries
- Home Energy Storage & Power Wall
- Nautical Applications
- Fishing, Boating Electronics
- Ice Fishing
- Recreational Vehicle
- Off-grid Life
- Deep Circulation Application

⚠Not Suitable for Starting Gasoline Engines

Golf Cart：

The motor power of 51.2V 100AH is less than the vehicle power of 5KW

The motor power of 38.4V 100AH is less than the vehicle power of 3.5KW

Warranty Exclusions:

The manufacturer's warranty does not cover:

Installation & Usage Related

1. Improper installation, use, maintenance, or servicing.
2. Insufficient or improper fastening of battery components.
3. Loose terminal connections.
4. Reverse polarity connections.
5. Electrical system malfunctions causing battery failure.
6. Use with unauthorized third-party products.
7. Product modifications without manufacturer's written approval.
8. Applications exceeding design specifications (including repeated engine starts or exceeding rated discharge current).

Configuration & Connections

1. Series connections exceeding 4 batteries (>204.8/153.6V systems).
2. Parallel connections exceeding 4 batteries.
3. Short circuits (accidental, intentional, or unintentional).
4. High resistance from corroded terminals, poor crimping, or undersized cables.

Physical & Environmental Damage

1. Impact, collision, or drop damage.
2. Improper storage conditions.
3. Force majeure events (fires, natural disasters, etc).

Maintenance & Charging

1. Improper charging (under/overcharging per manual specifications).
2. Batteries left uncharged for over 12 months.
3. Commercial applications with ⩾80% daily depth of discharge.
4. Failure to follow recommended charging protocols.

General Conditions

1. Any damage resulting from user negligence or failure to follow safety guidelines.
2. Consequences of product misuse or improper operation.

All warranty claims are subject to manufacturer's inspection and approval. Proper installation and maintenance as specified in the user manual are required to maintain warranty coverage.

Warranty and Returns

After-Sales Support

1. For any product support needs, please contact our Dumfume after-sales team directly through your order.
2. Our technical specialists will provide prompt assistance.

Battery Storage Guidelines

1. For optimal storage:
   Charge the battery to approximately 50% capacity.
   Disconnect all loads by removing the negative terminal connection.
   Store in a temperature-controlled environment (ideal: 25°C/77°F).
2. Storage Characteristics:
   Typical self-discharge rate: <3% per month (at 25°C/77°F).

Note: Discharge rates may increase significantly under extreme environmental conditions.

FAQS

Q: My battery shows 0V. What should I do?

A: A 0V reading typically indicates BMS protection activation. Follow these steps:

- Disconnect the battery from all loads.
- Measure the voltage with a multimeter:

  Above 47.2V (51.2V model)/35.4V (38.4V model)? Reconnect and resume use.

  Below 40V (51.2V model)/30V (38.4V model)? please contact our Dumfume after-sales team directly through your order.

Q: Should I charge the battery before first use?

A: Yes, always fully charge the battery before initial use.

⚠ Important: This also applies when connecting batteries in series or parallel.

Q: How do I check battery voltage?

A: Use a multimeter if no battery monitor is available. Connect probes to terminals for an accurate reading.

Q: Are there mounting restrictions?

A: Our batteries can be installed in any orientation. Ensure terminals are:

- Properly secured.
- Protected from accidental contact.

Thank You

Dear Valued Customer,

We sincerely appreciate your trust in Dumfume batteries. Your support inspires our team to continuously innovate and deliver cutting-edge battery solutions with uncompromising service excellence.

It's through your valuable feedback and the tireless commitment of our team that we're able to offer superior products at competitive prices, ensuring your complete satisfaction. Your confidence in our brand motivates us to keep raising the bar in energy solutions.

We take great pride in engineering these products and hope they meet your expectations. Should you need any assistance, our team remains at your service.

With gratitude,
The Dumfume Team

DUMFUME

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